Shahjalal University of Science and Technology, Sylhet

Department of Electrical and Electronic Engineering

EEE 0714 3229 Communication Engineering 1

Session: 2022-23

Term Tests 2 and 3 **Session: 2022-23** **Term Test 3 held on 10/11/26**

Term Test 2 (its questions 1 to 10 are 7(a) to 7(j))

[Communication – TT2 Questions]

7. (a) Advantages and disadvantages of FM over AM.

   (b) An audio frequency signal $10\sin 2\pi 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi 10^5 t$. Calculate:

   (i) Modulation index
   (ii) Sideband frequencies
   (iii) Amplitude of each sideband frequency
   (iv) Bandwidth required
   (v) Total power delivered to the load of 600 Ω
   (vi) Transmission efficiency

   (c) Explain switching modulators, specifically the ring modulator: circuit, working, and the case $m(t) = 0$ where the carrier is suppressed.

   (d) Compare AM (DSB-WC), DSB-SC and SSB-SC in modulation, demodulation, power efficiency and bandwidth.

   (e) Explain the effect of local oscillator phase error and frequency error in coherent detection of DSB-SC/SSB.

   (f) Disadvantages of Weaver's method of SSB generation.

   (g) What is VSB (vestigial sideband)? Why is it used?

   (h) Why is VSB-WC chosen for TV broadcasting?

   (i) Derive the narrowband FM ($\beta < 1$) expression.

   (j) Draw the NBFM block diagram.

Term Test 3 (its questions 1 to 3 are 8(a) to 8(c))

[Term Test 3, Time: 30min, Total Marks: 20]

8. (a) Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3MHz and $\Delta f = 20.48kHz$. A NBFM generator of $f_{c1} = 20kHz$ and $\Delta f = 5Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 50-kHz [*sic*: should be "500 kHz"] is readily available for frequency mixing. 10

   (b) Draw the block diagram of a Differential PSK (DPSK) modulator and demodulator and show how the I_k sequence [1010011100] changes in different parts of the block. 4

   (c) Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse stuffing TDM with no synchronization bits. Answer the following questions:

   a. What is the size of a frame in bits?

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b. What is the frame rate?
c. What is the duration of a frame?
d. What is the data rate? 6

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